

Keenan Eikenberry

✉ Keenan.J.Eikenberry@dartmouth.edu

🏠 <https://keenan-eikenberry.github.io/>

Research Interests

- 📖 Neural architectures (theory and applications)
- 📖 Applications of optimal transport to machine learning and statistical inference

Current Position

2023 – Present 📖 **Postdoctoral Research Associate**, Dartmouth College
Department of Mathematics, Hanover, NH
Supervisors: Yoonsang Lee and Anne Gelb

Education

- 2019 – 2023 📖 **Ph.D. in Applied Mathematics**, Arizona State University
Advisors: Doug Cochran and Shiwei Lan
Diss.: *Bayesian Inference for Markov Kernels Valued in Wasserstein Spaces*
- 2014 – 2016 📖 **M.A. in Mathematics**, Arizona State University
Advisors: John Quigg and Steven Kaliszewski
Thesis: *Functorial Results for C^* -algebras of Higher-Rank Graphs*
- 2009 – 2014 📖 **B.S. in Mathematics**, Arizona State University
B.A. in English Literature (minor in Philosophy), Arizona State University
Summa cum laude, GPA: 4.0

Papers and Code

Papers

- 1 **Eikenberry, K.**, Liu, L., & Lee, Y. (2026). Post-training augmentation invariance. *Transactions on Machine Learning Research*. Retrieved from <https://openreview.net/forum?id=Z4uUwU6zRe>
- 2 Eikenberry, S. E., Mancuso, M., Iboi, E., Phan, T., **Eikenberry, K.**, Kuang, Y., ... Gumel, A. B. (2020). To mask or not to mask: Modeling the potential for face mask use by the general public to curtail the covid-19 pandemic. *Infectious disease modelling*, 5, 293–308.

Code

- 1 Post-Training Augmentation Invariance. (2026).
https://github.com/keenan-eikenberry/augmentation_invariance.

Selected Talks

- 📖 Wasserstein Dependence and Representation Learning. Applied and Computational Mathematics Seminar, Dartmouth College. November 2024.
- 📖 Maximizing Multiview Wasserstein Dependence in Autoencoders. SIAM Mathematics of Data Science Conference, Atlanta, Georgia. October 2024.

Selected Talks (continued)

- Markov Kernels Valued in Wasserstein Spaces. Online CodEx Seminar. July 2023.
- Stacks of C^* -Categories. Great Plains Operator Theory Symposium, Texas A&M University. May 2019.
- Group Actions on Topological k -Graphs. C^* -Algebra Seminar, Arizona State University. November 2016.
- k -Graphs from Cayley Graphs. Great Plains Operator Theory Symposium, University of Illinois at Urbana-Champaign. May 2016.
- Functoriality for Higher-Rank Graph Algebras. C^* -Algebra Seminar, Arizona State University. March 2016.

Skills

Mathematics	■ Analysis, optimal transport, probability, category theory, partial differential equations, differential geometry, algebraic topology, abstract algebra, operator algebras
ML and AI	■ Neural architectures, theory of deep learning, self-supervised representation learning, optimal transport-based methods, scientific machine learning
Programming	■ Python (expert in PyTorch), Julia